(a) (b)

Figure 4: (a) Cessna T-41 A. (b) Cessna Citation X.

Table 1: Characteristics of the Cessna T-41 A and the Cessna Citation X shown in Fig.(4).

		Cessna T-41 A	Cessna Citation X
Wing span	[ft]	35.8	53.3
Wing area	[ft^2]	174	318
Normal gross weight	[lb]	2950	19815
Zero-lift drag coefficient	[−]	0.025	0.02
Stalling speed at sea-level	[kn]	43	86
Oswald efficiency factor	[−]	0.8	0.81
Power plant		One-piston engine of 230 hp at sea-level	Two-turbofan engine of 3650 lb thrust each at sea-level
Propeller efficiency	[−]	0.8	-

5 Steady Level Flight

Question 1

Consider the characteristics reported in Tab.(1) of a Cessna T-41 A (propeller engine) and a Cessna Citation X (jet engine) in level unaccelerated flight conditions. The airplanes are shown in Fig.(4). In addition, consider that the power of the Cessna T-41's propeller as a function of the altitude may be approximated using the formula

$$P_{A_{\max}}(h_G) = \left(\frac{\rho(h_G)}{\rho_{\mathrm{SL}}} - \frac{1-\rho(h_G)/\rho_{\mathrm{SL}}}{7.55}\right) P_{A_{\max,\mathrm{SL}}}, \tag{2}$$

whereas the thrust of the Cessna Citation X may be approximated as

$$T_{A_{\max}}(h_G) = \left(\frac{\rho(h_G)}{\rho_{\mathrm{SL}}}\right) T_{A_{\max,\mathrm{SL}}}. \tag{3}$$

Then

- Plot the required thrust T_R and the required power P_R at sea-level as a function of the velocity V and identify the minimum-thrust velocity and the minimum-power velocity.
- Plot the required thrust T_R and the required power P_R at $h_G = 5000\,\mathrm{m}$, and briefly discuss the differences with the results obtained at sea-level.
- Compute the maximum velocity at $h_G = 0$ and $h_G = 5000\,\mathrm{m}$ for the two airplanes.